EX-MA CABLE EXCAVATOR · VERSION B (MINIMAL, NO PVC CONDUIT)

CUT PLAN FOR YOUR BOARDS

Three 7/16″ OSB panels (2 × 4 ft) and one 5/8″ particle-board panel (2 × 2 ft). Every size below is worked out for those thicknesses; the box and the sandbox both stand straight on the ground (no shim).

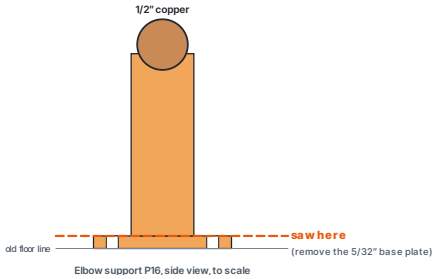
Particle board
sandbox floor W17 — use it whole, no cuts

2 OSB panels
everything else: 1704 sq in (74 % of the 2 panels)

1 OSB panel spare
keep it for re-cuts or a lid

WHAT CHANGES BECAUSE OF YOUR BOARDS

- **7/16″ OSB instead of 12 mm (1/32″ thinner).** Outside sizes and every hole position stay as designed; pieces that fit between others come out up to 1/16″ longer. All sizes in this plan already include that.
- **5/8″ floor, no shim.** The sandbox floor top sits 5/32″ higher than the design. The **pedestal walls and sandbox walls are 5/32″ shorter** to match, so the pedestal top, the turret and both conduit fittings stay at their design heights and still line up with the control box. Hole heights in those walls are already corrected.
- **Elbow support (printed P16):** it stands on the sandbox floor, so it is now 5/32″ too tall. Saw off its 5/32″ base plate flush with the post (drawing on the right) and fix the post with one #6 × 1-1/4″ screw up through the floor into its centre (1/8″ pilot) plus a few drops of CA or epoxy. Nothing else printed is affected.
- **Spacer stack** is now 5 × 7/16″: the chamber between the fittings is 3 1/8″ long — the PTFE tubes' bow needs about 11/16″ radius, still above the 5/8″ minimum DERIVED. The arm's slew axis moves 2 3/8″ toward the box.



Elbow support P16, side view, to scale

LENGTHS FOR THIS BUILD DERIVED

8-32 through-rods (4)	**3 1/16″**
Boom / stick / bucket rope	**49 5/8″ / 66 7/8″ / 80 11/16″**
PTFE stick_hi / stick_lo	**23 1/16″ / 23 1/2″**
PTFE bucket_hi / bucket_lo	**29 9/16″ / 29 15/16″**
Slew sleeve A / B	**10 9/16″ / 13 3/16″**
Slew rope A / B	**29 7/8″ / 32 5/8″**
Lever axle / spool axle / 3 handles (5/16″ rod)	**7-5/16″ / 7-3/8″ / 5-11/16″ each**

Ropes and tubes are shorter than the manual by the 2 7/16″ the conduit run lost (arm ropes twice). Cut ropes 3/4″ long and trim when you crimp.

WORKING WITH THESE MATERIALS ASSUMPTION

PARTICLE BOARD AND SAND. Particle board swells and crumbles if it gets damp. Seal both faces and all four edges of the floor (paint, poly or a coat of wood glue thinned with water) before assembly, and keep the sand dry.

OSB EDGES. Screws hold less well in OSB edges than in plywood. Drill a 3/32″ pilot for every #6 screw, keep screws 1/2″ from corners, and glue every joint.

VERSION B · YOUR BOARDS

CUT LIST

Sizes for 7/16″ OSB and the 5/8″ particle-board floor. Tick each piece as you cut it and pencil its ID on it, with an arrow to its **bottom** edge.

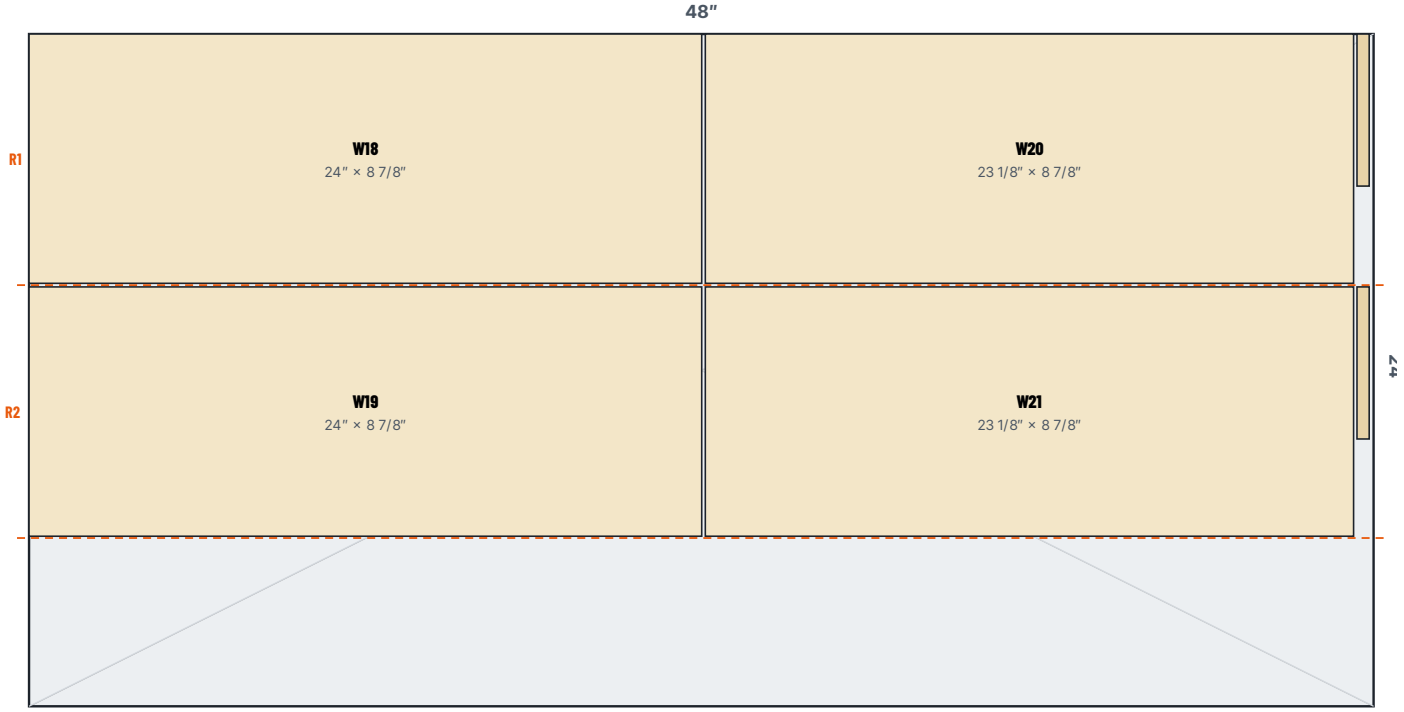
	ID	Piece	Cut size (in)	mm	Holes	sq in
☐	W01	box floor	**14 1/4″ × 8 7/16″**	362 × 214	1	120
☐	W02	box front	**14 1/4″ × 5 1/4″**	362 × 134	7	75
☐	W03	box rear	**14 1/4″ × 5 1/4″**	362 × 134	—	75
☐	W04	box right side	**7 9/16″ × 5 1/4″**	192 × 134	—	40
☐	W05	box left side	**7 9/16″ × 5 1/4″**	192 × 134	—	40
☐	W06	box top (removable)	**14 1/4″ × 8 7/16″**	362 × 214	4	120
☐	W07	axle bearing block left	**4″ × 2 3/8″**	101 × 60	1	9
☐	W08	axle bearing block right	**4″ × 2 3/8″**	101 × 60	1	9
☐	W09	pedestal top	**7 3/8″ × 7 1/16″**	188 × 180	5	52
☐	W10	pedestal wall -u (conduit)	**7 1/16″ × 6 7/8″**	180 × 175	5	49
☐	W11	pedestal wall +u (removable access)	**7 1/16″ × 6 7/8″**	180 × 175	—	49
☐	W12	pedestal wall -v	**6 7/8″ × 6 1/2″**	175 × 166	—	45
☐	W13	pedestal wall +v	**6 7/8″ × 6 1/2″**	175 × 166	—	45
☐	W14	pedestal shelf	**6 3/16″ × 5 7/16″**	158 × 138	1	34
☐	W15	pedestal shelf cleat -v	**5 7/16″ × 7/16″**	138 × 11	—	2
☐	W16	pedestal shelf cleat +v	**5 7/16″ × 7/16″**	138 × 11	—	2
☐	W17	sandbox floor	**24″ × 24″**	610 × 610	—	576
☐	W18	sandbox wall -u (box side)	**24″ × 8 7/8″**	610 × 226	7	214
☐	W19	sandbox wall +u	**24″ × 8 7/8″**	610 × 226	—	214
☐	W20	sandbox wall -v	**23 1/8″ × 8 7/8″**	588 × 226	—	206
☐	W21	sandbox wall +v	**23 1/8″ × 8 7/8″**	588 × 226	—	206
☐	W22	spacer frame 1	**7 1/16″ × 6 7/8″**	180 × 175	1	49
☐	W23	spacer frame 2	**7 1/16″ × 6 7/8″**	180 × 175	1	49

VERSION B · YOUR BOARDS

BOARD LAYOUTS AND CUT ORDER

Rip the strips first (orange dashed lines, R1, R2 …), then crosscut each strip. Drawn to scale with a 1/8″ kerf; grey is offcut.

OSB PANEL 1 — 2 × 4 FT, 7/16″



Thin strips not labelled on the drawing: W15, W16 — see the cut order.

R1: rip a strip **8 7/8″** wide; crosscut: 24″ → **W18**; 23 1/8″ → **W20**; 7/16″ → **W15**.

R2: rip a strip **8 7/8″** wide; crosscut: 24″ → **W19**; 23 1/8″ → **W21**; 7/16″ → **W16**.

VERSION B · YOUR BOARDS

BOARD LAYOUTS AND CUT ORDER (CONTINUED)

OSB PANEL 2 — 2 × 4 FT, 7/16″



R1: rip a strip **8 7/16″** wide; crosscut: 14 1/4″ → **W01**; 14 1/4″ → **W06**; 7 3/8″ → **W09**; 7 1/16″ → **W10**; 4″ → **W07** + **W08** (rip that piece into 2 3/8″ / 2 3/8″).

R2: rip a strip **6 7/8″** wide; crosscut: 7 1/16″ → **W11**; 7 1/16″ → **W22**; 7 1/16″ → **W23**; 6 7/8″ → **W12**; 6 7/8″ → **W13**; 6 3/16″ → **W14**.

R3: rip a strip **5 1/4″** wide; crosscut: 14 1/4″ → **W02**; 14 1/4″ → **W03**; 7 9/16″ → **W04**; 7 9/16″ → **W05**.

OSB PANEL 3 — SPARE

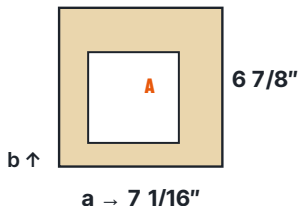
Not needed. Keep it for a re-cut.

PARTICLE BOARD — 2 × 2 FT, 5/8″

W17 sandbox floor: use the whole panel, no cuts (24″ × 24″; the design is 1/64″ bigger, which does not matter). Seal it before building on it.

DRILLING — SPACER FRAMES W22 + W23

W22 + W23 spacer frames (×2, identical)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

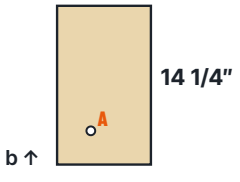
	Hole / bit	a / b (in)	mm	For
A	**window 3 15/16″ square**: corners at a 1 1/4″ – 5 3/16″, b 1″ – 4 15/16″ 100 mm	**3 1/4″ / 3″**	82 / 76	window (drill 3/8″ in each corner, then jigsaw)

VERSION B · YOUR BOARDS

DRILLING

Measure **a** along the bottom edge and **b** up the side, from the lower-left corner as drawn. The bold size is the drill bit. In OSB, clamp a scrap under the panel so the holes do not tear out.

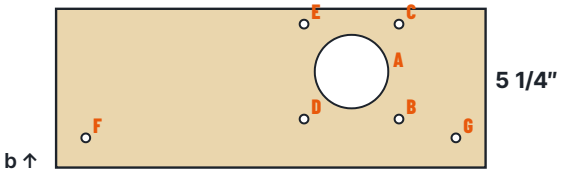
W01 box floor



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**11/32"** Ø8.5 mm	**3 1/16" / 3 1/16"**	77 / 77	slew spool axle (nut + washer in a Ø22 × 7 counterbore from below)

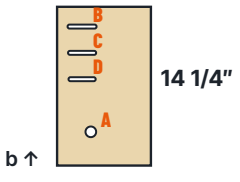
W02 box front



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16" hole saw** Ø62.0 mm	**9 13/16" / 3 3/16"**	249 / 81	2in conduit
B	**3/16"** Ø4.6 mm	**11 3/8" / 1 5/8"**	289 / 41	8-32 through-rod
C	**3/16"** Ø4.6 mm	**11 3/8" / 4 3/4"**	289 / 121	8-32 through-rod
D	**3/16"** Ø4.6 mm	**8 1/4" / 1 5/8"**	209 / 41	8-32 through-rod
E	**3/16"** Ø4.6 mm	**8 1/4" / 4 3/4"**	209 / 121	8-32 through-rod
F	**3/16"** Ø4.6 mm	**1" / 1"**	25 / 25	box-to-sandbox bolt (M3 × 30)
G	**3/16"** Ø4.6 mm	**13 1/4" / 1"**	337 / 25	box-to-sandbox bolt (M3 × 30)

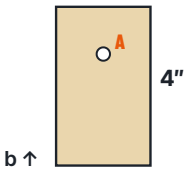
W06 box top (removable)



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1" spade / Forstner** Ø24.4 mm	**3 1/16" / 3 1/16"**	77 / 77	spool tube bearing
B	**slot 2 5/8"**: 23/64" holes at a 1 1/8" & 3 3/8", saw between 66.1 mm	**2 1/4" / 12 1/2"**	57 / 317	boom lever slot = hard stop (42 deg swing)
C	**slot 2 1/2"**: 23/64" holes at a 1 3/16" & 3 5/16", saw between 64.1 mm	**2 1/4" / 10 1/8"**	57 / 257	stick lever slot = hard stop (41 deg swing)
D	**slot 2 7/16"**: 23/64" holes at a 1 3/16" & 3 5/16", saw between 61.8 mm	**2 1/4" / 7 3/4"**	57 / 197	bucket lever slot = hard stop (40 deg swing)

W07 + W08 axle bearing blocks (×2, identical)



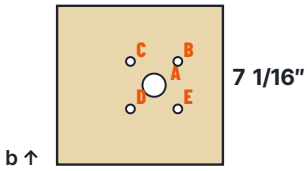
a from the −u edge, b from the bottom edge — seen from the −v side

	Hole / bit	a / b (in)	mm	For
A	**11/32"** Ø8.5 mm	**1 3/16" / 2 13/16"**	30 / 71	5/16in lever axle

VERSION B · YOUR BOARDS

DRILLING (CONTINUED)

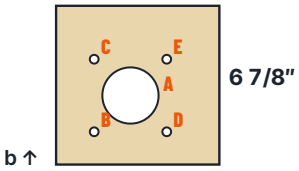
W09 pedestal top



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1″ spade / Forstner** Ø26.2 mm	**4 5/16″ / 3 9/16″**	110 / 90	PEX turret tube clearance
B	**1/8″** Ø3.0 mm	**5 3/8″ / 4 5/8″**	137 / 117	pilot, #8 screw from the base
C	**1/8″** Ø3.0 mm	**3 1/4″ / 4 5/8″**	83 / 117	pilot, #8 screw from the base
D	**1/8″** Ø3.0 mm	**3 1/4″ / 2 1/2″**	83 / 63	pilot, #8 screw from the base
E	**1/8″** Ø3.0 mm	**5 3/8″ / 2 1/2″**	137 / 63	pilot, #8 screw from the base

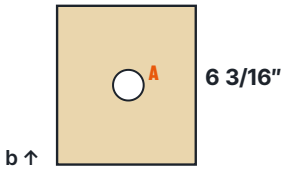
W10 pedestal wall -u (conduit)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**3 1/4″ / 3″**	82 / 76	2in conduit
B	**3/16″** Ø4.6 mm	**1 5/8″ / 1 7/16″**	42 / 36	8-32 through-rod
C	**3/16″** Ø4.6 mm	**1 5/8″ / 4 9/16″**	42 / 116	8-32 through-rod
D	**3/16″** Ø4.6 mm	**4 13/16″ / 1 7/16″**	122 / 36	8-32 through-rod
E	**3/16″** Ø4.6 mm	**4 13/16″ / 4 9/16″**	122 / 116	8-32 through-rod

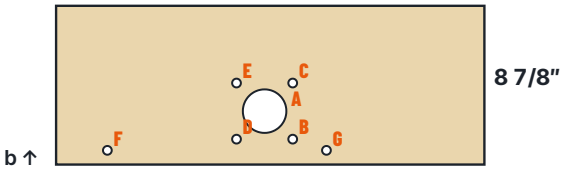
W14 pedestal shelf



a from the −u edge, b from the −v (right) edge — seen from above

	Hole / bit	a / b (in)	mm	For
A	**1-1/4″ spade / Forstner** Ø30.4 mm	**2 13/16″ / 3 1/8″**	71 / 79	slew-tube-bushing

W18 sandbox wall -u (box side)



a from the −v (right) edge, b from the bottom edge — seen from the +u side

	Hole / bit	a / b (in)	mm	For
A	**2-7/16″ hole saw** Ø62.0 mm	**11 11/16″ / 3″**	297 / 76	2in conduit (seal with silicone)
B	**3/16″** Ø4.6 mm	**13 1/4″ / 1 7/16″**	337 / 36	8-32 through-rod
C	**3/16″** Ø4.6 mm	**13 1/4″ / 4 9/16″**	337 / 116	8-32 through-rod
D	**3/16″** Ø4.6 mm	**10 1/8″ / 1 7/16″**	257 / 36	8-32 through-rod
E	**3/16″** Ø4.6 mm	**10 1/8″ / 4 9/16″**	257 / 116	8-32 through-rod
F	**3/16″** Ø4.6 mm	**2 7/8″ / 13/16″**	73 / 20	bolt from the control box
G	**3/16″** Ø4.6 mm	**15 3/16″ / 13/16″**	385 / 20	bolt from the control box